

Logic & Form Are All You Need

逻辑与形式即一切

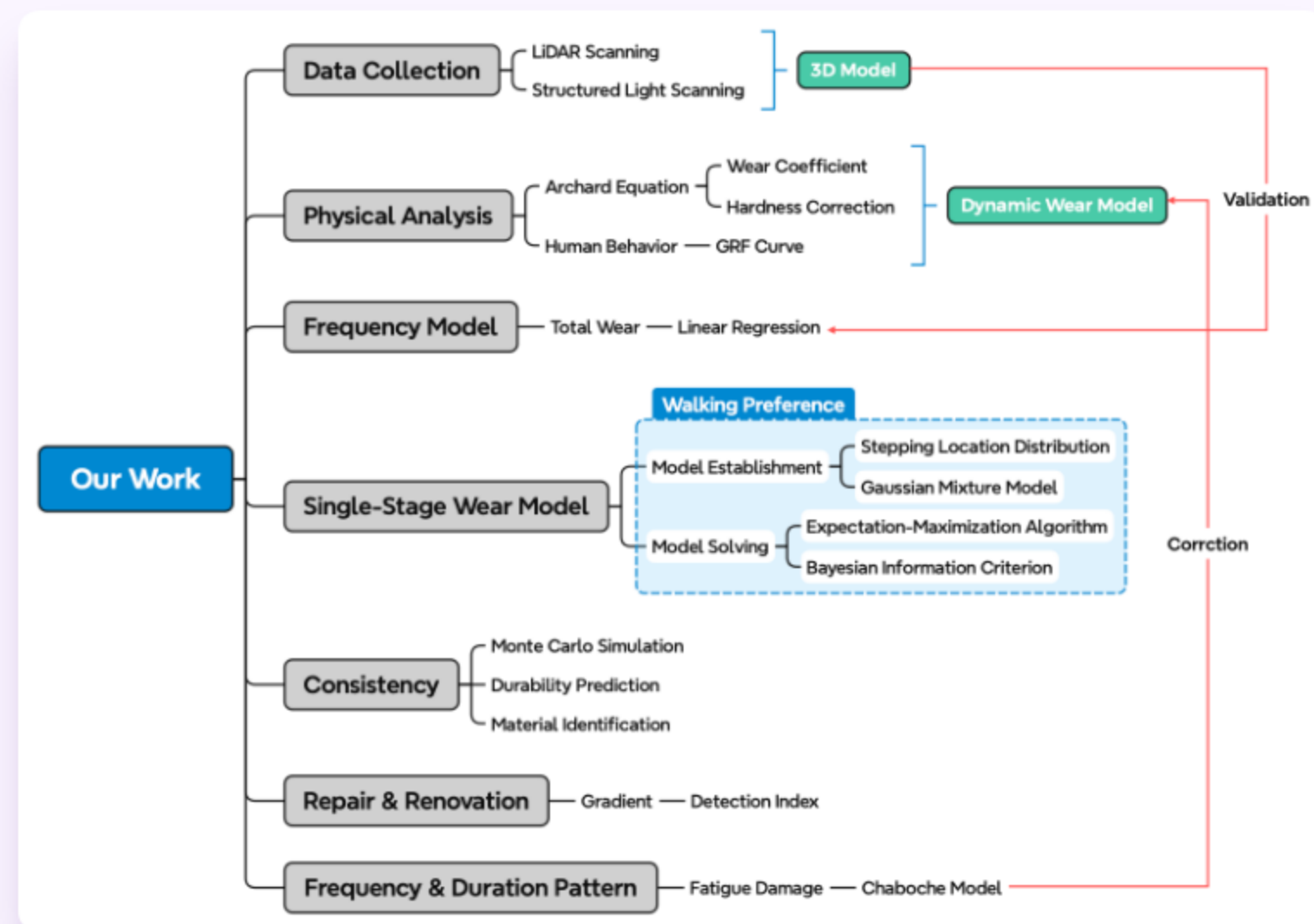
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文章概述

Step Chronicles



1. 数据采集

2. 模型构建

3. 模型评估

Step Chronicles: A Multiscale Model Linking Wear Patterns and Temporal Use in Ancient Staircases

Summary

Staircase wear is a complex phenomenon influenced by various factors such as material properties, frequency of use, and environmental conditions. However, understanding this wear is crucial in archaeology, providing valuable insights into ancient societies, their daily activities, and technological advancements. To aid archaeological studies, we developed models to simulate staircase wear, offering a more precise understanding of how wear is produced.

In Task 1, we establish a model based on **Archard's Wear Law** and **Human Behavior Analysis** to address the usage frequency and volume of the wear of staircases. By analyzing wear depth $W(x, y)$ and fitting **Ground Reaction Forces** data, the model estimates the quantity of the wear and the frequency of staircase usage f_{use} . Besides, validation through 3D scanning data further confirm the model's accuracy.

In Task 2, we employ a probabilistic model based on and **Gaussian Mixture Model** and **Bayesian Information Criterion** to analyze the direction and usage pattern of staircases. The model evaluates the preference of the direction by comparing n_{up} and n_{down} , which are statistically computable parameters in the model by **Expectation-Maximization Algorithm**. The number of people using the stairs simultaneously, which is related to the number of mixture Gaussian distribution, can be accessed by minimizing Bayesian Information Criterion.

In Task 3, we develop a model combining **Monte Carlo simulations** and **Smoothness Detection** to verify wear consistency and material origin. The model evaluates wear depth $W(x, y)$ and surface gradient $S(x, y)$ to detect disruptions in natural wear patterns, indicating repairs or renovations. The model also simulate wear progression during usage and predict the remaining service life before reaching the safety limit. Apart from that, we conduct **wear coefficient analysis and statistical tests** to identify the source material of the stairs, combining quantitative methods with archaeological evidence for accurate verification. To access the frequency and duration pattern, the fatigue damage of stairs under different stepping patterns is analyzed using the **Chaboche model**, showing that wear is accelerated by short-duration high-frequency stepping compared to long-duration low-frequency stepping.

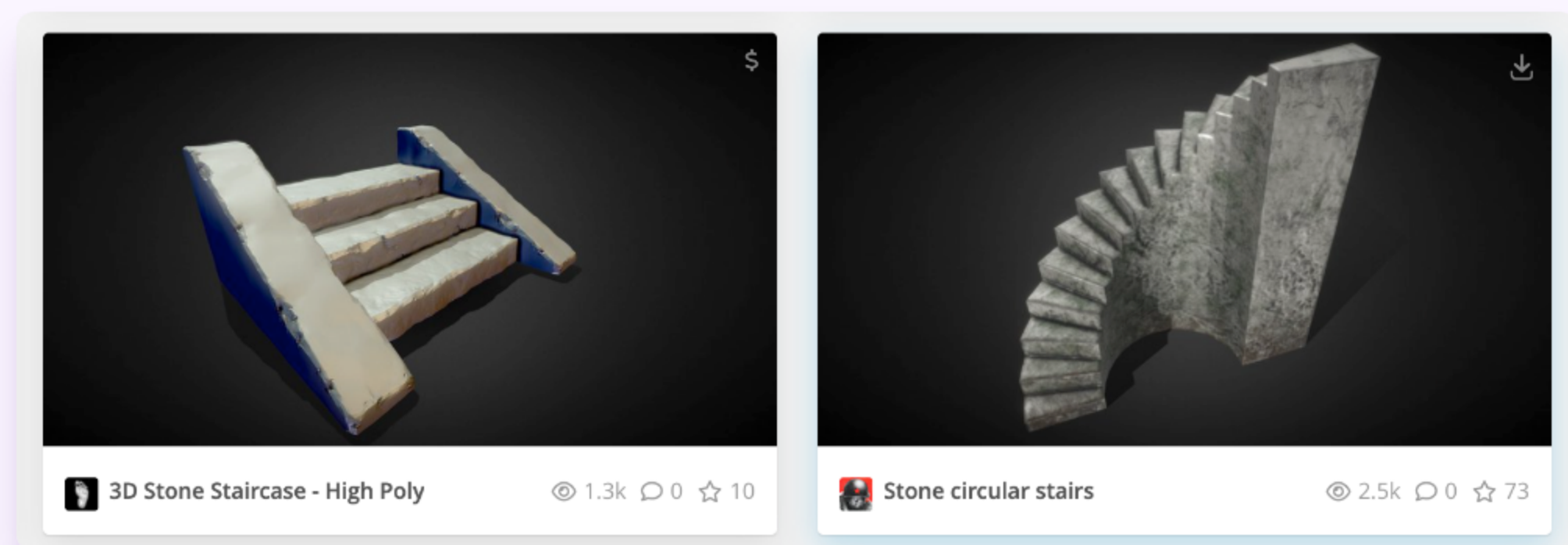
Finally, the model reliably predicts staircase wear but has limitations, and future improvements will focus on material adaptations, dynamic environmental factors, and multi-scale modeling.

Keywords: Stair Wear; Gaussian Distribution; Monte Carlo Simulation; Fatigue Damage

数据采集

A 题很难找到数据

游戏资产中的台阶模型



3D 建模软件处理，转化为点云图，转化成便于处理的数据

- Blender
- Maya
- ZBrush

物理分析

Archard 磨损定律

$$V_{\text{wear}} = k \cdot \frac{F \cdot L}{H}$$

经典 Archard 方程描述了磨损体积 V 与法向载荷 F 、滑动距离 L 及材料硬度 H 之间的关系，对其进行修正：

$$V_{\text{wear}} = \alpha \cdot \frac{h}{H} \cdot g(M)$$

式中， α 为材料磨损系数， h 为人类特征参数， $g(M)$ 为硬度修正系数（温度）

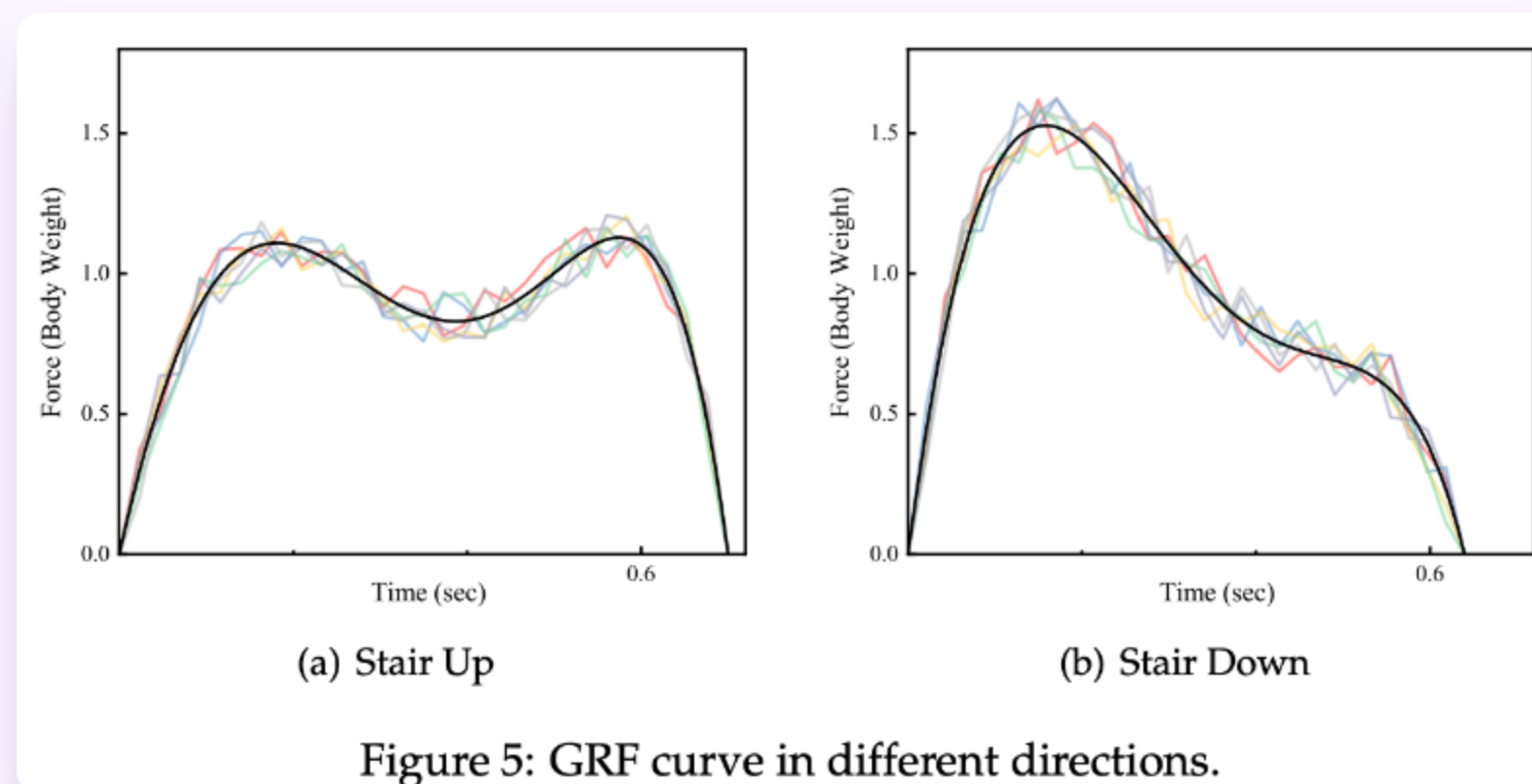
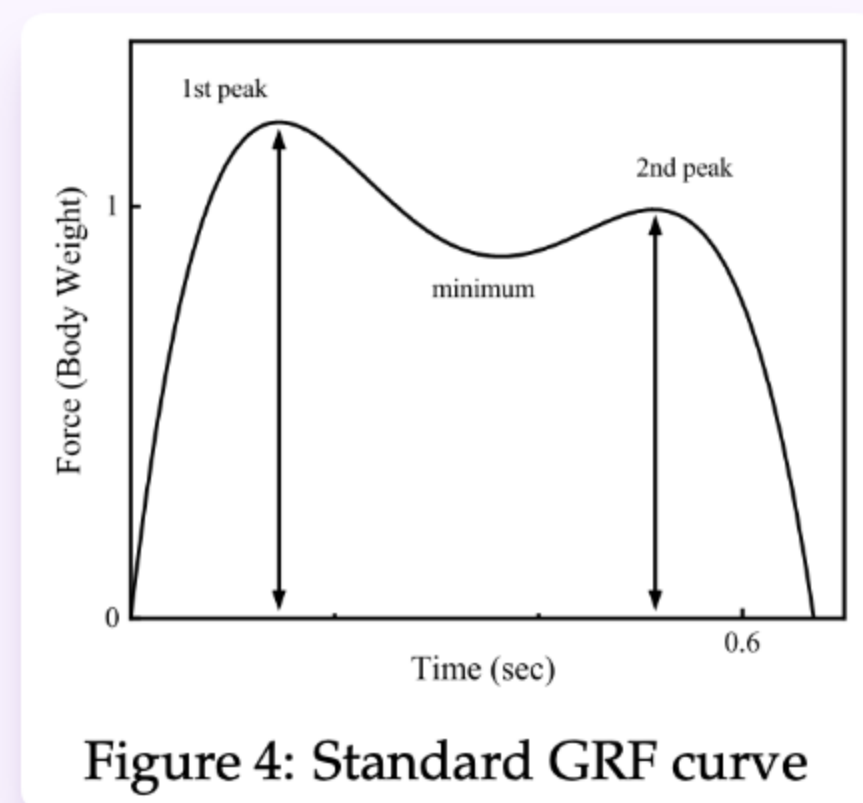
$$h = \int \rho(w) \cdot P(w) \cdot C(w) \cdot L(w) dw$$

式中， ρ 为体重 PDF， P 为压强， C 为接触面积

物理分析

GRF Curve

通过 Stüssi 和 Debrunner 的研究，单次步行的垂直 GRF 曲线呈典型的“M”型

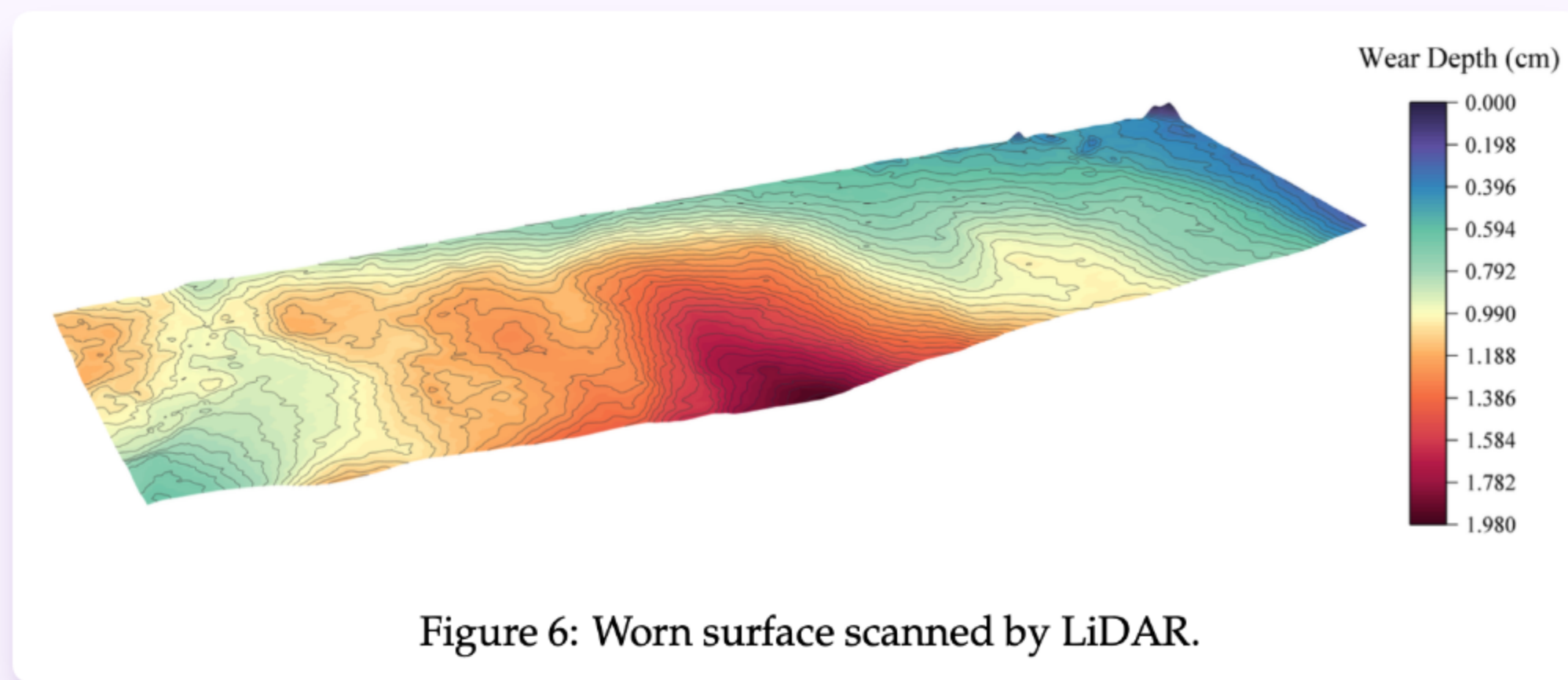


上下楼的 GRF 曲线存在显著差异，拟合这些曲线并积分，得到上楼与下楼的磨损比

$$\frac{w_{\text{up}}}{w_{\text{down}}} = \frac{0.4335}{0.3804} \approx 1.14$$

频率分析

3D 扫描得到总磨损体积



通过磨损体积得到频率

$$V_{\text{wear,tot}} = V_{\text{wear}} \cdot f_{\text{use}} \cdot T$$

物理模型计算得到的磨损与实际磨损进行误差比对

单级磨损模型

分析行走偏好

概率模型构建

将单级台阶表面视为平面区域 $\mathcal{D}$ ，磨损深度 $W(x, y)$ 与踩踏次数 $N(x, y)$ 成正比

$$W(x, y) = w_{\text{up}} \cdot N_{\text{up}}(x, y) + w_{\text{down}} \cdot N_{\text{down}}(x, y)$$

将踩踏位置视为随机变量，服从概率分布，根据大数定律，磨损深度分布反映了脚落点的概率分布

$$W(x, y) = w_{\text{up}} n_{\text{up}} \cdot p_{\text{up}}(x, y) + w_{\text{down}} n_{\text{down}} \cdot p_{\text{down}}(x, y)$$

高斯混合模型 (GMM): 假设踩踏位置服从二维联合高斯分布。因为人有双脚且可能多人并行，分布是多个高斯分布的混合。

$$p_{\text{up}}(x, y) = \frac{1}{k} \sum_{i=1}^k \mathcal{N}(\mathbf{x} | \mu_i, \Sigma_i)$$

概率模型构建

$$W(x, y) = \frac{1}{k} \left(w_{\text{up}} n_{\text{up}} \sum_{i=1}^k \mathcal{N}(x, y | \mu_i, \Sigma_i) + w_{\text{down}} n_{\text{down}} \sum_{i=1}^k \mathcal{N}(x, y | \mu_i, \Sigma_i) \right)$$

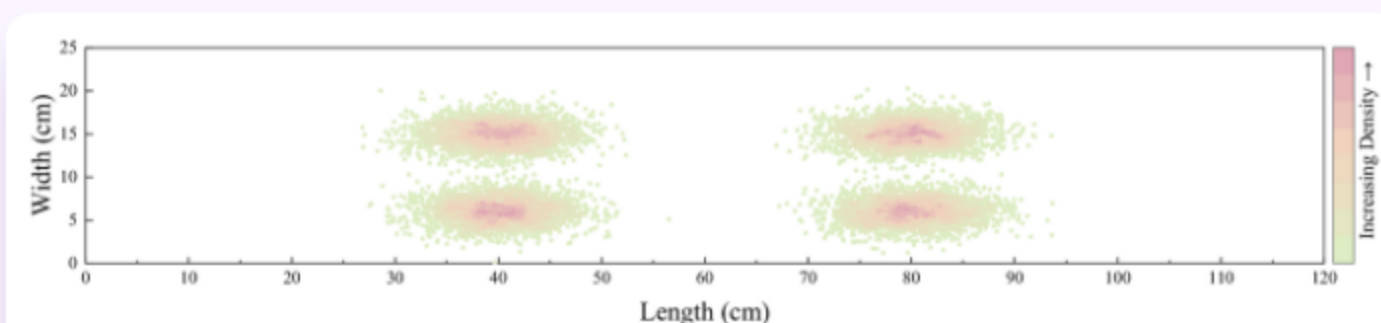


Figure 7: The distribution of the location where feet stepping on.

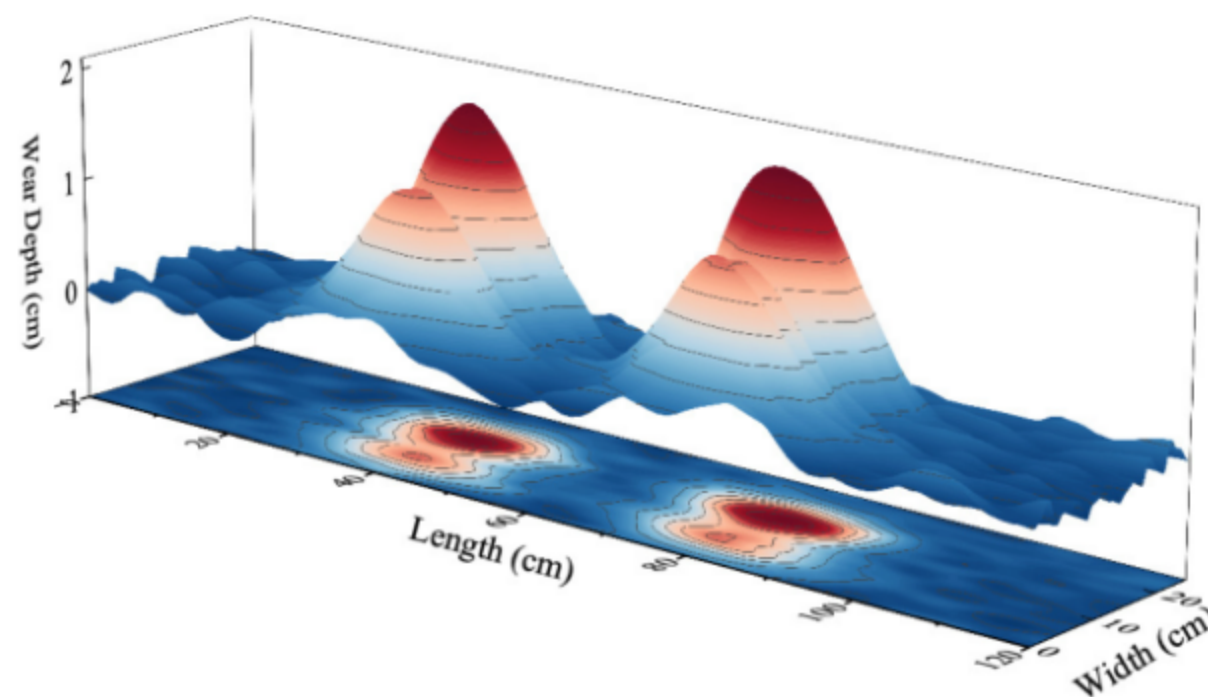


Figure 8: The distribution of the wear depth

模型求解

将磨损深度函数 $W(x, y)$ 进行归一化处理, 使其变成一个概率密度函数 $p_W(x, y)$

$$p_W(x, y) = \frac{W(x, y)}{\iint_{\mathcal{D}} W(x, y) \, dx \, dy} = \sum_{i=1}^{2k} \pi_i \cdot \mathcal{N}(x, y | \mu_i, \Sigma_i)$$

$$\pi_i = \begin{cases} \frac{w_{\text{up}}(n_{\text{up}} + \varepsilon_i)}{k \iint_{\mathcal{D}} W(x, y) \, dx \, dy} & 1 \leq i \leq k, \\ \frac{w_{\text{down}}(n_{\text{down}} + \varepsilon_i)}{k \iint_{\mathcal{D}} W(x, y) \, dx \, dy} & k < i \leq 2k \end{cases}$$

模型求解

EM 算法 (Expectation-Maximization)

- 第一步：初始化 (Initialization) 随机猜测参数 π_i (权重)、 μ_i (中心位置)、 Σ_i (形状范围) 的初始值
- 第二步：E 步 (Expectation Step) 计算 $\gamma_{m,i}$ ，即对于每一个数据点 x_m ，计算它属于第 i 个高斯分布 (即第 i 个脚印簇) 的概率是多少
- 第三步：M 步 (Maximization Step) 根据 E 步算出的概率，更新参数
 - 更新权重 π_i
 - 更新中心位置 μ_i
 - 更新协方差 Σ_i
- 第四步：检查收敛计算对数似然函数 l ，如果数值变化很小 (趋于稳定)，就停止迭代，否则重复 E 步和 M 步

模型求解

EM 算法算出参数 π_i 后，结合总的磨损体积（对 $W(x, y)$ 积分），就可以算出上楼和下楼的人数

$$n_{\text{up}} + \varepsilon_i = \frac{\pi_i k}{w_{\text{up}}} \iint_{\mathcal{D}} W(x, y) \, dx \, dy$$

$$n_{\text{down}} + \varepsilon_i = \frac{\pi_i k}{w_{\text{down}}} \iint_{\mathcal{D}} W(x, y) \, dx \, dy$$

判断方向

- 如果 $n_{\text{up}} > n_{\text{down}}$ ，则主要流向是上楼
- 如果 $n_{\text{up}} < n_{\text{down}}$ ，则主要流向是下楼

模型求解

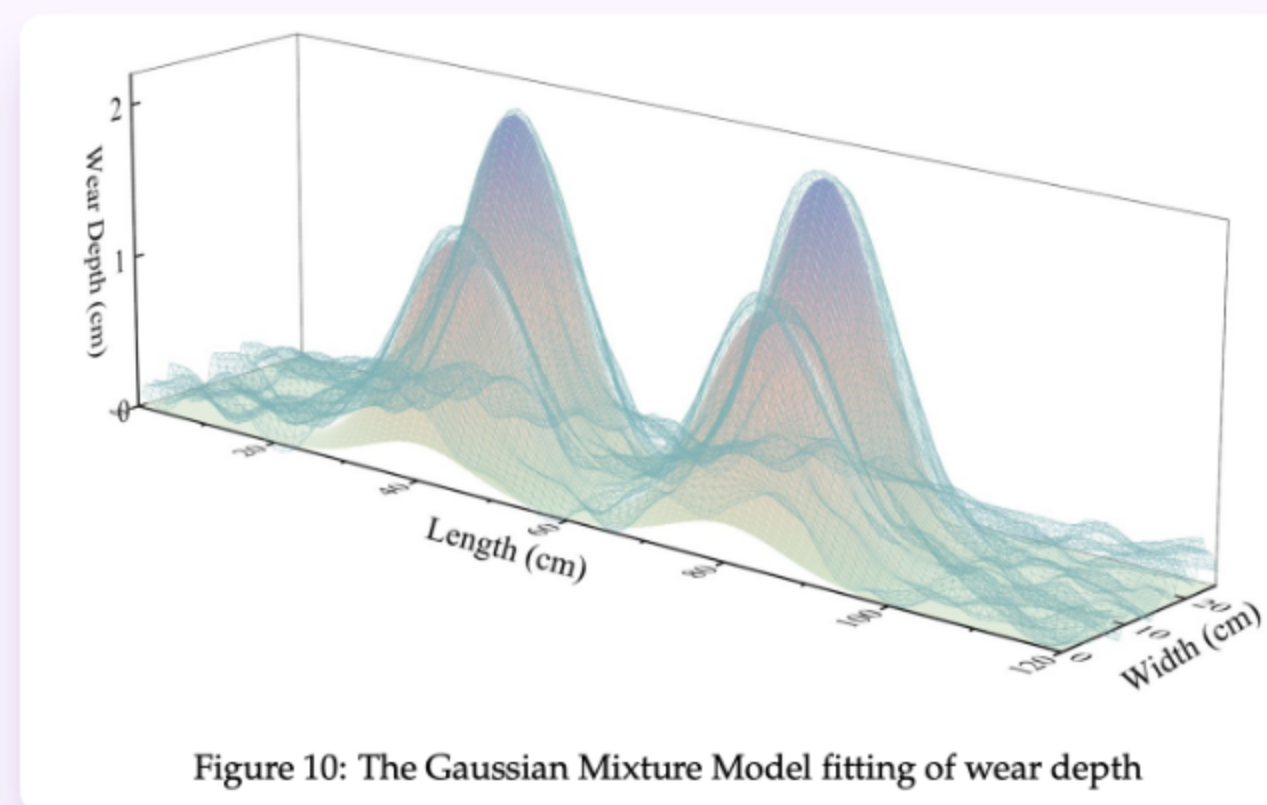
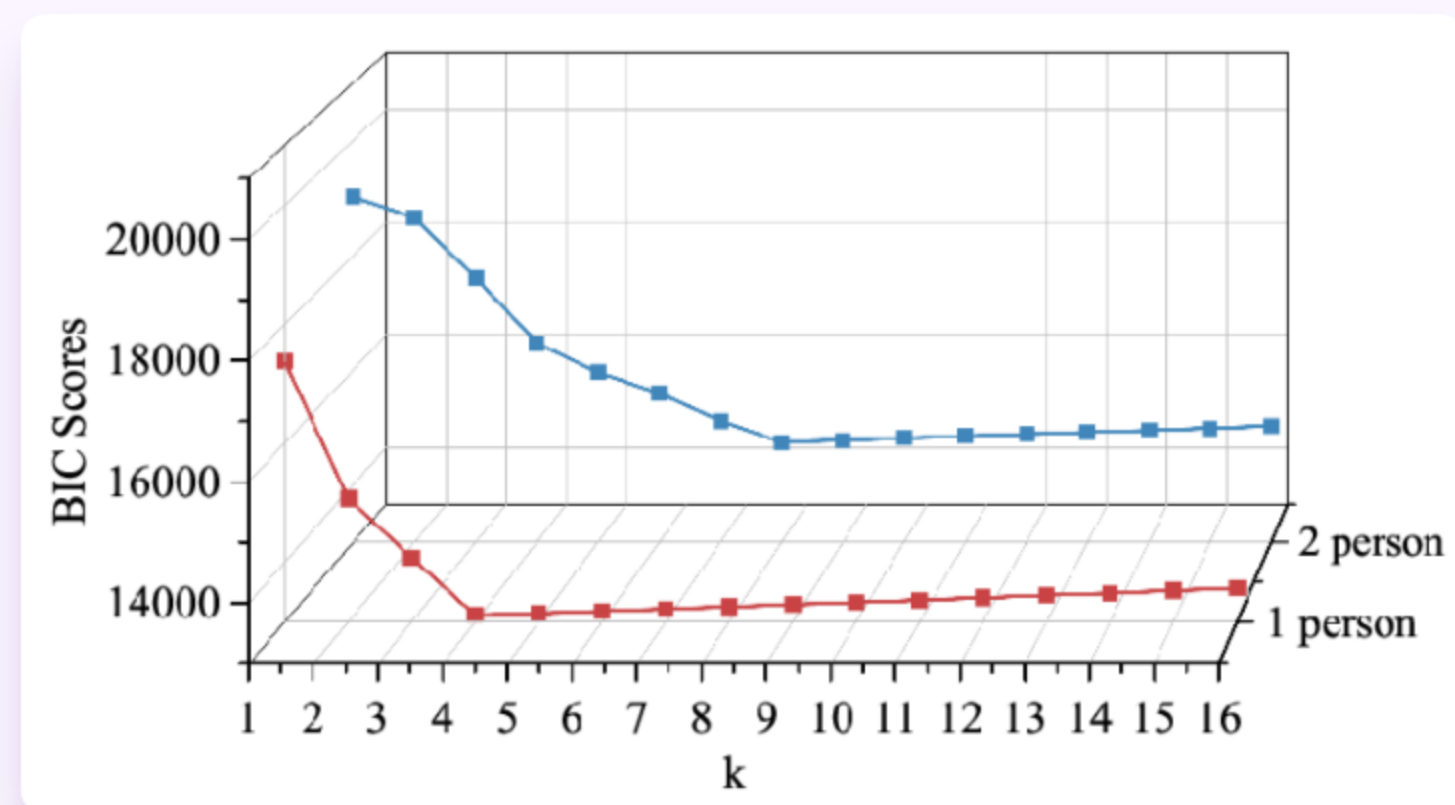
贝叶斯信息准则 (BIC)

用 BIC 准则确定到底有几个簇最合理 (以此判断是几个人并排走)

$BIC = p \ln M - 2l$, 值越小越好

设定 k 从 1 到 10, 分别运行 EM 算法, 计算 BIC 值

选择 BIC 值最小的那个 k , 因为人有两只脚, 所以同时使用楼梯的人数 = $k/4$



磨损一致性比对

通过对比磨损程度与建筑年代、历史记录来验证一致性

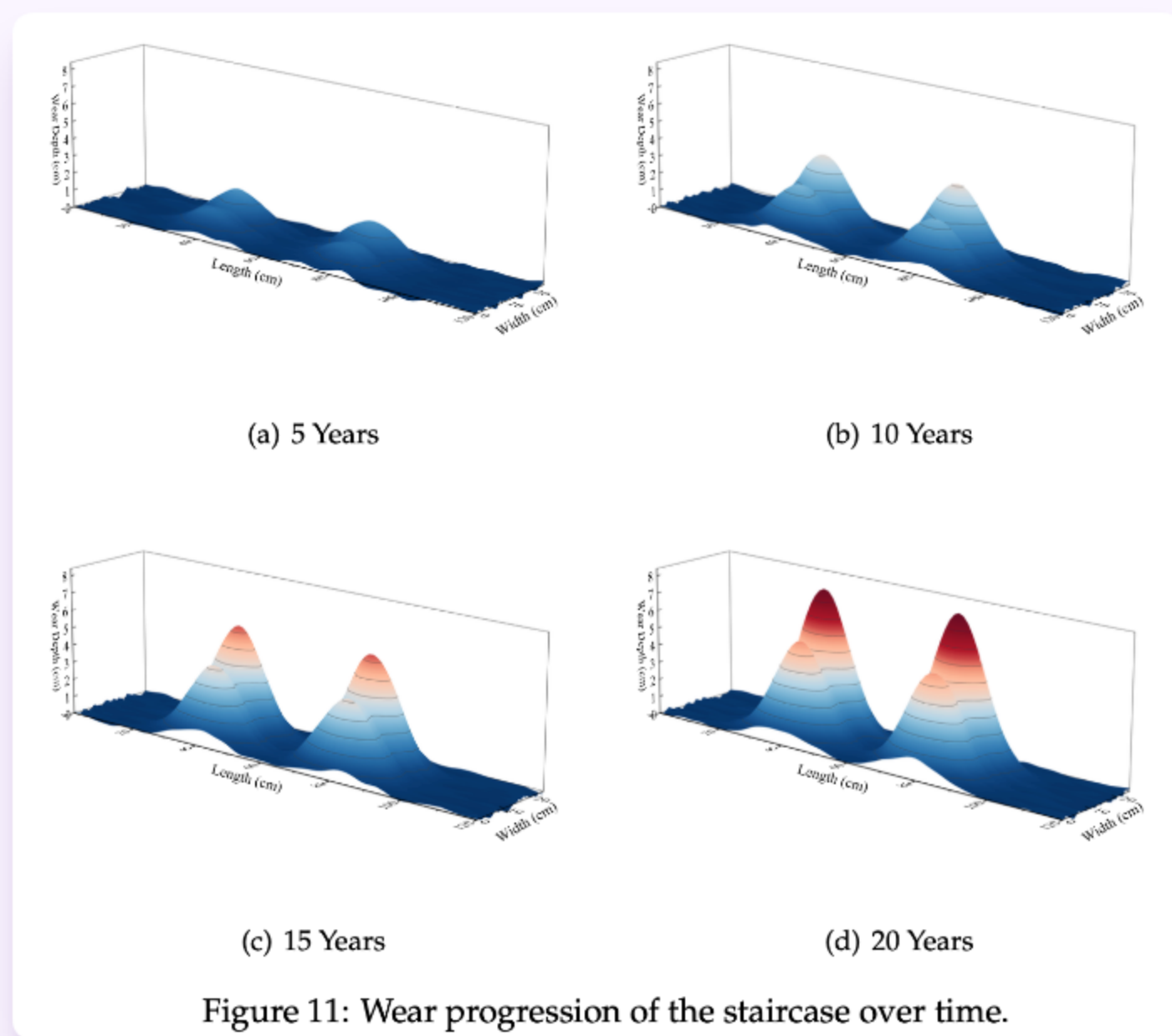
利用蒙特卡洛方法模拟随机行走过程，生成理论磨损图

```
1  wear ← [l, w];
2  for time ← 0 to T do
3      for person ← 1 to N do
4          weight ← NormalDistribution(mu_w, sigma_w);
5          parallel ← NormalDistribution(mu_n, sigma_n);
6          if Random() < p_up then
7              direction ← UP;
8          else
9              direction ← DOWN;
10         end
11         normal_force ← GRF(weight, direction);
12         sliding_distance ← L(weight, direction);
13         h ← normal_force * sliding_distance;
14         (x, y) ← PositionGenerator(direction, parallel);
15         wear[x, y] ← wear[x, y] + alpha * (h / H) * g;
16     end
17 end
```

楼梯耐久性预测

基于模型预测剩余寿命

阈值：当最大磨损深度达到台阶高度时，视为不可用



修复与翻新检测

梯度检测

自然磨损产生渐变梯度，而修复会引入梯度的突变

定义表面梯度 $S(x, y)$ 和修复检测指数 $R(x, y)$ (局部梯度的偏差)

$$S(x, y) = \sqrt{\left(\frac{\partial W}{\partial x}\right)^2 + \left(\frac{\partial W}{\partial y}\right)^2}$$

$$R(x, y) = \left| S(x, y) - \bar{S}_{\text{local}}(x, y) \right|$$

R 值大表示可能存在修复痕迹

材料来源识别

通过摩擦学测试测定疑似产地材料的磨损系数 α ，并计算现有楼梯的有效磨损系数 α_e

- 获取标准数据，获取疑似来源地（比如附近的采石场或伐木场）材料的磨损数据，进行摩擦学测试，测定这些“标准样本”的磨损系数
- 提取样本特征，反推楼梯数据对于考古现场的楼梯，不进行破坏性测试，结合现场观测到的磨损量，反向计算出楼梯材料的有效磨损系数
- 使用了统计学中的假设检验

t-检验

$$t = \frac{K - K_e}{s/\sqrt{n}}$$

Kolmogorov-Smirnov 检验

$$D = \sup_x |F_1(x) - F_2(x)|$$

人流频率和持续时间

疲劳损伤检定

- 短时高频 (Short-duration high-frequency) : 比如短时间内大量人群通过, 产生快速、高幅度的冲击
- 长时低频 (Long-duration low-frequency) : 比如长时间内零星的人流, 应力较小但持续累积

Chaboche 模型

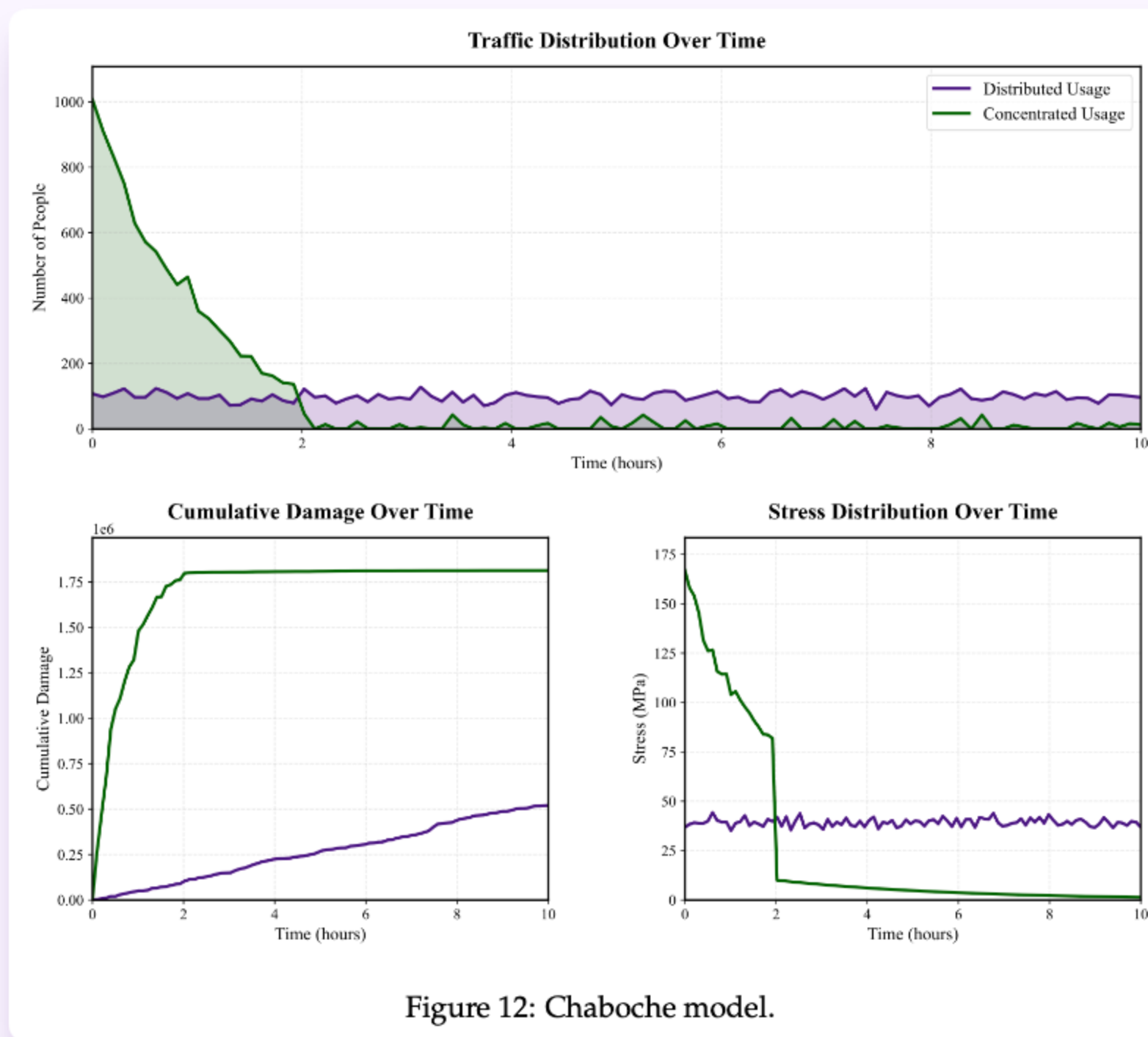
$$f = \sqrt{\frac{3}{2}(\mathbf{s} - \mathbf{X}) : (\mathbf{s} - \mathbf{X})} - \sigma_y = 0$$

$$\dot{\mathbf{X}}_i = \frac{2}{3}C_i\dot{\boldsymbol{\epsilon}}_p - \gamma_i\mathbf{X}_i\dot{p} \quad (i = 1, 2, \dots, N)$$

$$\dot{D} = \left(\frac{Y}{S}\right)^s \cdot \dot{p} \cdot \langle \sigma_H \rangle^m$$

人流频率和持续时间

疲劳损伤检定



模型评估

Strengths & Limitations & Future Directions

■ 模型优点

- 理论结合实际：将物理定律（Archard 磨损定律）与人的随机行走行为（概率统计）结合了起来，平衡了理论深度和实际应用
- 处理不确定性：使用了蒙特卡洛模拟，这让模型能够很好地处理现实中人流和环境的各种随机变化
- 准确度高：通过在花岗岩楼梯上的实际验证，模型的预测误差仅为 4.89%，证明在控制条件下的可靠性

■ 模型局限性

- 假设过于理想化：模型假设楼梯材料是完全均匀的，且人的走路习惯也是理想化的，现实中，石材可能有杂质，人的行为也千奇百怪
- 环境因素被简化：虽然考虑了环境影响（如湿度、温度），但只是用一个修正系数 $g(M)$ 来概括，现实中，环境是动态变化的（比如冻融循环），这部分在模型中没有完全体现
- 修复检测的短板：用“梯度分析法”来检测楼梯是否修补过，对于那些磨损特别不规则的楼梯，可能难以区分是自然磨损还是修补痕迹

模型评估

Strengths & Limitations & Future Directions

- 未来的改进方向
 - 考虑特殊材质：让模型适应非均匀材料（比如有纹理的木头或层状岩石），根据纹理方向调整磨损系数
 - 加入动态环境：让修正系数 $g(M)$ 变成随时间变化的函数，比如能反映季节性的湿度变化
 - 微观与宏观结合：将宏观的磨损预测与微观层面的摩擦模拟结合起来，更细致地研究磨损颗粒是如何产生的

文章撰写

Rhetoric

摘要

最重要的部分

- 段落结构清晰
- 突出主要使用的数理工具 / 算法模型
- 反复打磨语言

标题

生动创新，点题

Problem Chosen
A

2025
MCM/ICM
Summary Sheet

Team Control Number
2501567

Step Chronicles: A Multiscale Model Linking Wear Patterns and Temporal Use in Ancient Staircases

Summary

Staircase wear is a complex phenomenon influenced by various factors such as material properties, frequency of use, and environmental conditions. However, understanding this wear is crucial in archaeology, providing valuable insights into ancient societies, their daily activities, and technological advancements. To aid archaeological studies, we developed models to simulate staircase wear, offering a more precise understanding of how wear is produced.

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Keywords: Stair Wear; Gaussian Distribution; Monte Carlo Simulation; Fatigue Damage

文章撰写

Rhetoric

1. 人称 (Person)

- 严禁使用第一人称单数 "I"
- 推荐使用第三人称
- 慎用但可以使用第一人称复数 "We"

2. 时态 (Tense)

- 一般现在时 (Present Tense): 用于描述不受时间限制的事实、你的模型结构、图表内容以及结论。
 - Example: "The simulation results indicate a strong correlation..." (结果表明...)
- 一般过去时 (Past Tense): 用于描述你已经做过的具体工作、实验过程或引用的过往研究。
 - Example: "Smith (2019) proposed a similar approach." (Smith曾提出...)
- ~~一般将来时 (Future Tense)~~: 仅在“模型优缺点”或“未来工作 (Future Work)”部分使用，描述如果时间允许还可以做什么。
 - Example: "In future studies, we will consider the impact of weather..."

文章撰写

Rhetoric

问题重述 (Restatement of the Problem)

不要复制粘贴原题！用自己的语言重新阐述问题背景 and 任务，体现你对问题的理解。

模型假设 (Assumptions)

Assumption -> Justification.

符号说明 (Notations)

使用表格，一列符号，一列定义，一列单位（如有）

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文章撰写

Rhetoric

模型建立与求解 (Model Building & Solution)

- 公式要编号，方便文中引用 ("As shown in Eq. (1)...")
- 逻辑流畅，不要突然扔出一个公式，要解释公式中每个变量的含义以及物理/数学意义
- 图文并茂

优缺点评价 (Strengths & Weaknesses)

承认模型的局限性（例如：没考虑极端天气，数据量不足），优点不要只写“模型准确”，要具体

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工具

Tools

L^AT_EX

重中之重，熟悉 L^AT_EX 语法，环境使用，不要造轮子，常见排版问题

- GitHub 上的 mcmthesis 模板
- physics 包
- `\begin{equation}` 替代 `$$`
- `\autoref` 管理图表公式引用
- BibLaTeX 管理文献 (Google Scholar)

Overleaf 团队协作

工具

Tools

润色&翻译

把 Gemini 改成人话，检查语法错误，专有名词使用

常用工具：

- Quillbot
- Grammarly
- Turnitin

提交前仔细核对，语法、排版、格式等等

工具

Tools

作图（形式）

尽量使用矢量图，PDF 形式插入

常用工具：

- Origin
- PowerPoint
- Adobe Illustrator
- Visio
- MATLAB / Python
- Xmind

Python 预处理数据 -> Origin 绘图

作图 重要性高于实际结论!!! 好看的图是大的加分项

尽量不要使用 AI 绘图，AI 生成图片注意字符清晰度，是否有错字漏字的情况

合理分配工作量



工具

Tools

模型（逻辑）

使用 AI 给出大致思路，再逐个击破，注意 AI 上下文限制

AI 可以解决大部分问题，但要注重逻辑的展开，自己要充分理解 AI 生成的内容，模型的建立要有理有据

STFW

各种问题，无论是排版、绘图等，善用搜索引擎解决问题，不要过于依赖 AI，谨防 AI 幻觉

- Google
- Google Scholar
- Z-Library

赛中

队友合作

提前大致分配好工作，安装需要的软件

开赛时分头整理好题目，做好选题工作

管理好个人情绪，尊重他人想法，切忌意见不合闹矛盾

后勤准备

建议线下，住酒店民宿/学校

准备物资，吃好喝好，合理作息

Thanks

Presented by Junhan XU, Shurui XU